Outliers and the Shape of Data

Name _____ Date _____ Calculator allowed Version 1 Scan to practise this topic online

KEY WORDS outlier *(tau tawhiti)* · mean *(toharite)* · median *(tau waenga)* · range *(whānuitanga)* · symmetrical

A Spotting outliers ●○○

Remember: An outlier is a value that is much larger or much smaller than the rest of the data.

1. Write the outlier in each set of data.

a) 12, 15, 14, 13, 48, 16 _____

b) 7, 8, 6, 9, 7, 1 _____

c) 150, 154, 149, 152, 98, 151, 155 _____

d) 2.1, 2.4, 2.2, 9.6, 2.3 _____

e) 33, 35, 30, 34, 31, 5, 32 _____

f) 60, 64, 61, 63, 62, 120 _____

B What an outlier does ●●○

Watch out: An outlier is not always a mistake. Check where it came from before you leave it out.

2. The ages, in years, of the people at a birthday party are: 11, 12, 12, 13, 12, 11, 13, 45

	Mean	Median	Range
With all the values			
Without the outlier			

a) Which value is the outlier?

b) Complete the table. Round the means to 1 decimal place.

c) Which measure does the outlier change the most? Which does it change the least?

3. The times, in seconds, that 7 students took to run 100 m are: 15, 16, 14, 17, 16, 15, 33

	Mean	Median	Range
With all the values			
Without the outlier			

a) Which value is the outlier?

b) Complete the table. Round the means to 1 decimal place.

c) Which measure does the outlier change the most? Which does it change the least?

4. A thermometer reading of 91°C is written in a list of outdoor temperatures that are all between 15°C and 22°C. Suggest how this outlier could have happened.

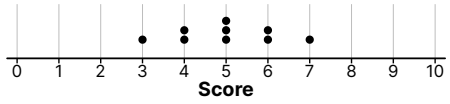
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C Symmetrical and skewed data ●●○

Remember: Symmetrical data looks about the same on both sides of the middle. Skewed data has a longer tail on one side: a tail to the right is a positive skew, and a tail to the left is a negative skew.

5. Describe the shape of each dot plot: symmetrical, skewed to the right or skewed to the left.

a)



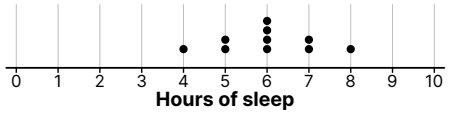
b)



c)



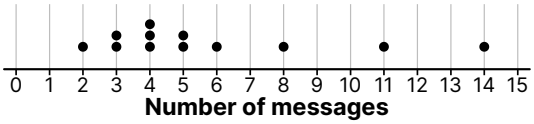
d)



D Shape, centre and spread ●●●

Remember: To describe a set of data, give its shape, its centre (the median) and its spread (the range), and mention any outliers.

6. The dot plot shows the number of text messages sent by each student in a group in one day.

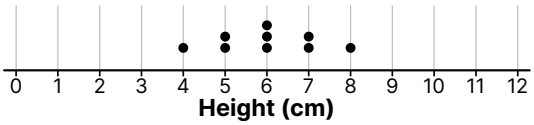


a) Describe the shape of the data.

b) Find the median (the centre).

c) Find the range (the spread).

7. The dot plot shows the heights, in centimetres, of the seedlings in a tray.



a) Describe the shape of the data.

b) Find the median (the centre).

c) Find the range (the spread).

Extension

8. The mean of the six numbers 20, 22, 21, 23, 24, 22 is 22. A seventh number, 78, is added. Find the new mean.

9. House prices in a street are mostly between $600,000 and $700,000, but one house sold for $3 million. A news story wants to give a "typical" price. Should it use the mean or the median? Explain.

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1-1 PRIVATE TUTORS

STATISTICS

Outliers and the Shape of Data

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS outlier *(tau tawhiti)* · mean *(toharite)* · median *(tau waenga)* · range *(whānuitanga)* · symmetrical

A Spotting outliers ●○○

Remember: An outlier is a value that is much larger or much smaller than the rest of the data.

1. Write the outlier in each set of data.

a) 11, 14, 13, 15, 52, 12 _____

b) 8, 9, 7, 9, 8, 2 _____

c) 160, 164, 159, 162, 105, 161, 165 _____

d) 3.1, 3.4, 3.2, 8.9, 3.3 _____

e) 43, 45, 40, 44, 41, 9, 42 _____

f) 70, 74, 71, 73, 72, 140 _____

B What an outlier does ●●○

Watch out: An outlier is not always a mistake. Check where it came from before you leave it out.

2. The ages, in years, of the people at a birthday party are: 12, 13, 13, 14, 13, 12, 14, 52

	Mean	Median	Range
With all the values			
Without the outlier			

a) Which value is the outlier?

b) Complete the table. Round the means to 1 decimal place.

c) Which measure does the outlier change the most? Which does it change the least?

3. The times, in seconds, that 7 students took to run 100 m are: 16, 17, 15, 18, 17, 16, 36

	Mean	Median	Range
With all the values			
Without the outlier			

a) Which value is the outlier?

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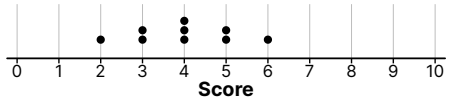
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C Symmetrical and skewed data ●●○

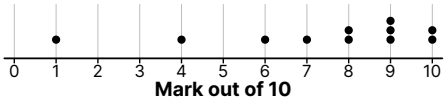
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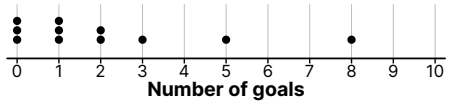
a)



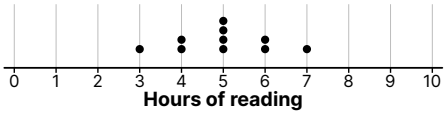
b)



c)



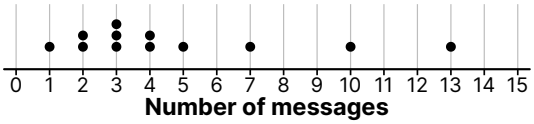
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D Shape, centre and spread ●●●

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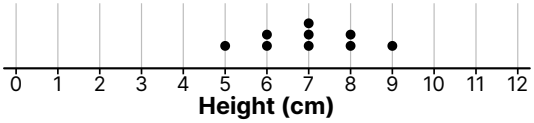


a) Describe the shape of the data.

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7. The dot plot shows the heights, in centimetres, of the seedlings in a tray.



a) Describe the shape of the data.

b) Find the median (the centre).

c) Find the range (the spread).

Extension

8. The mean of the six numbers 30, 32, 31, 33, 34, 32 is 32. A seventh number, 95, is added. Find the new mean.

9. House prices in a street are mostly between $600,000 and $700,000, but one house sold for $3 million. A news story wants to give a "typical" price. Should it use the mean or the median? Explain.

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